

DISTRIBUTION-AWARE NEURAL ADDITIVE MODELS: ROBUST INTERPRETABLE DEEP LEARNING WITH FEATURE SELECTION

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ABSTRACT

Neural Additive Models (NAMs) have gained traction as an interpretable class of deep learning models, offering a favorable trade-off between transparency and accuracy. However, the effectiveness of NAMs in real-world applications is severely constrained by complex noise structures arising from their reliance on classical loss functions that lack inherent robustness. To address the above issues, we propose a novel learning framework, the **Distribution-Aware Neural Additive Model** (DANAM). The proposed model is rooted in maximum likelihood estimation for adaptive error modeling, enabling it to capture diverse noise structures directly from data without imposing restrictive distributional assumptions. Furthermore, to improve predictive performance and interpretability, we incorporate group sparsity regularization into DANAM, which can automatically select the most informative input features. Experiments verify the effectiveness and robustness of the proposed DANAM against several state-of-the-art approaches, especially under challenging noisy data conditions. The implementation code can be found at <https://github.com/zx1ml/DANAM>.

Index Terms— Neural additive model, feature selection, group sparsity, robustness, kernel density estimation

1. INTRODUCTION

Deep learning has achieved remarkable success on complex prediction tasks, yet its interpretability remains limited compared to classical approaches. Neural Additive Models (NAMs) [1] address this limitation by decomposing predictions into an additive collection of univariate subnetworks, each associated with a single input feature. This structure enables NAMs to achieve strong performance across diverse tasks [2] and to be increasingly adopted in high-stakes domains [3, 4], including healthcare and biology [5, 6]. Recent work extends NAMs along two main directions: one focuses on improving computational efficiency through parameter sharing [7] and high-level concept representations [8], which reduce the overhead of training and maintaining separate

subnetworks; the other aims to enhance interpretability by incorporating first-order and higher-order feature interactions beyond the univariate setting, enabling more flexible and data-driven structure discovery [9, 10, 11].

While NAM architectures have seen substantial progress in recent years, most existing methods still rely on classical loss functions that remain highly sensitive to data contamination. In classification settings, logistic loss is commonly used, and its statistical efficiency often depends on the strong assumption of perfectly reliable labels. However, abnormal data (e.g., outliers and mislabeled samples) are inevitable in practice, which may introduce systematic bias into the estimates and destabilize the decision boundaries. This limitation motivates the adoption of robust likelihood-based alternatives [12, 13], which mitigate the influence of corrupted labels while preserving predictive accuracy.

To address these challenges in data modeling, a valuable approach is to employ Maximum Likelihood Estimation (MLE), which offers a flexible and theoretically sound alternative in statistical machine learning [14, 15]. MLE aims to pinpoint the parameter values that maximize a likelihood function, which represents the probability of the observed data under a specified statistical model. Building upon these principles, we formulate new additive models, called **Distribution-Aware Neural Additive Models** (DANAM), for the noise distribution-aware setting. The main contributions of this work are highlighted below.

- **Robust Model Architecture:** We propose a new class of additive models (called DANAM), that integrates kernel density estimation with a data-adaptive maximum likelihood objective. Without imposing explicit distributional assumptions, DANAM flexibly handles diverse noise structures, including outliers, mislabeled samples, and imbalanced class distributions.
- **Feature Selection:** DANAM incorporates a sparsity-inducing regularization at the feature level, allowing the model to focus on variables that carry genuine predictive signal. This mechanism yields more interpretable representations by emphasizing the inputs most relevant to the task.

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- **Empirical Validation:** Experiments on real-world classification datasets show that DANAM exhibits robustness against outliers and mislabeled samples, and achieves improved predictive accuracy.

2. PRELIMINARIES

This section reviews the basic setup of the nonparametric model and presents the architecture of DANAM.

2.1. Problem Formulation

Let $\mathcal{X} \subset \mathbb{R}^p$ be a compact input space and $\mathcal{Y} \subset \mathbb{R}$ be an output set. We consider a nonparametric model:

$$y = f^*(\mathbf{x}) + \epsilon, \quad (1)$$

where the input $\mathbf{x} = (x^1, \dots, x^p)^\top \in \mathcal{X}$, the response $y \in \mathcal{Y}$, and ϵ is the random noise independent of $\mathbf{x}$. The primary objective is to estimate the true underlying function f^* based on a finite set of observations $\mathbf{z} := \{(\mathbf{x}_i, y_i)\}_{i=1}^n \subset \mathcal{X} \times \mathcal{Y}$, where each sample is drawn independently from the distribution determined by (1).

For classification tasks under the assumption that the labels follow a Bernoulli distribution, a common choice is the logistic loss estimator

$$\hat{f}_{\log} = \arg \min_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n \log \left(1 + \exp \left(-y_i \cdot f(\mathbf{x}_i) \right) \right), \quad (2)$$

where $y_i \in \{-1, 1\}$ is the ground-truth label, and $\mathcal{F}$ is the hypothesis space. Nevertheless, the logistic loss function in (2), which performs well under clean data, can break down in the presence of outliers or label noise, leading to unstable and biased estimates. In fact, samples with large loss values, such as misclassified outliers or borderline examples, can dominate the objective, diminishing the influence of well-classified samples that provide more reliable supervision. This limitation underscores the need for robust estimators that are adaptive to the underlying error distribution.

Motivated by [13], we adopt a robust estimation framework that models the empirical error distribution by formulating the problem as MLE on the residuals. The objective is to maximize the joint log-likelihood of the residuals:

$$\hat{f}_{\text{MLE}} = \arg \max_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n \log h(\epsilon_i), \quad (3)$$

where h is the probability density function of the residual ϵ_i . Since h is usually unknown in practice, we employ kernel density estimation via a kernel function K , such as the Gaussian kernel or an asymmetric kernel, to estimate the density function:

$$\hat{h}(\epsilon_k) = \frac{1}{nu} \sum_{i=1}^n K\left(\frac{\epsilon_k - \epsilon_i}{u}\right), \quad (4)$$

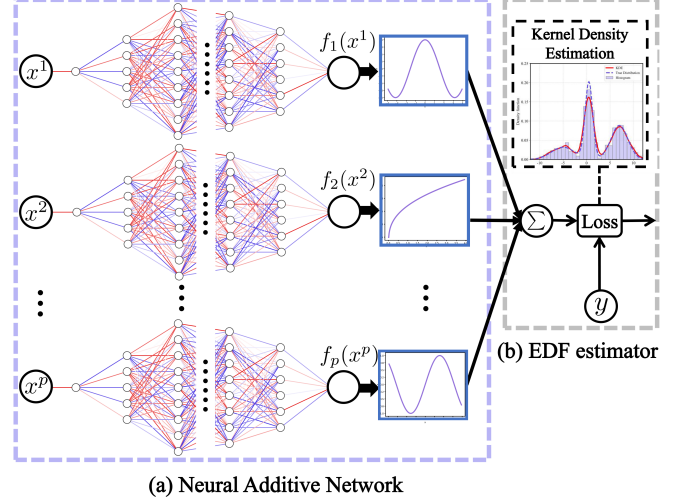


Fig. 1. The architecture of DANAM. Each input variable is modeled as a group, with a dedicated neural network learning its corresponding shape function. The overall loss is derived from kernel density estimation of the residuals and backpropagated to update the model parameters.

where u is a bandwidth.

Replacing the unknown true density $h(\cdot)$ in the MLE objective with its empirical estimate $\hat{h}$, we obtain the Estimated Density Function (EDF) estimator. That is,

$$\hat{f}_{\text{EDF}} = \arg \min_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n \left(-\log \hat{h}(\epsilon_i) \right). \quad (5)$$

This data-adaptive loss function incorporates the underlying data structure without relying on restrictive a priori assumptions about the noise distribution [13, 16].

2.2. Distribution-Aware Neural Additive Models

As natural extensions of linear models, additive models characterize the relationship between covariates and the response variable as a sum of univariate functions [17, 18]. The hypothesis function in additive models is defined as $f(\mathbf{x}) = \sum_{j=1}^p f_j(x^j)$, where $f_j : \mathbb{R} \rightarrow \mathbb{R}$ maps the j -th feature x^j of the input $\mathbf{x}$ to its corresponding contribution. This decomposition mitigates the curse of dimensionality by simplifying high-dimensional estimation and simultaneously enhances interpretability by making feature contributions explicit [19].

Recently, NAMs [1] have extended classical additive models by employing a small, dedicated neural network to parameterize each component function f_j . This architecture combines the expressiveness of DNNs with the inherent intelligibility of generalized additive models [17]. The additive hypothesis space $\mathcal{H}$ employed by NAMs is then defined as

$$\mathcal{H} = \left\{ f : f = \sum_{j=1}^p f_j, f_j \in \mathcal{H}_j \right\}, \quad (6)$$

where $\mathcal{H}_j$ is the neural hypothesis space. Building on this framework, we next formulate the optimization problem of DANAM, as shown in Figure 1, as follows:

$$\min_{\Theta} \frac{1}{n} \sum_{i=1}^n \mathcal{L}(\mathbf{x}_i, y_i; \Theta) + \lambda \alpha(\Theta), \quad (7)$$

where $\Theta := [\theta_1, \dots, \theta_p]$ denotes the set of all trainable parameters in DANAM, $\mathcal{L}(\mathbf{x}_i, y_i; \Theta)$ is the task-specific loss function and $\alpha(\Theta)$ represents group sparsity regularization. Notably, the group structure imposed on subnetworks underlies the feature selection mechanism in DANAM: it imposes a structured penalty on Θ that ensures that the entries within each group are jointly set to either zero or non-zero.

For classification tasks, we instantiate the loss function $\mathcal{L}$ by applying the EDF principle. Specifically, the density estimator is applied to the probabilistic prediction error, defined as the deviation from perfect confidence in the ground-truth class. This leads to the objective:

$$\mathcal{L}(\mathbf{x}_i, y_i; \Theta) = -\log \left[\hat{h} \left(1 - \sigma \left(y_i \cdot \sum_{j=1}^p f_j^{\theta_j}(\mathbf{x}_i^j) \right) \right) \right], \quad (8)$$

where $\sigma(\cdot)$ is the sigmoid function, and θ_j denotes the parameters associated with the j -th subnetwork. This formulation encourages the predictive distribution to concentrate its mass near unit confidence for the correct label.

3. OPTIMIZATION ALGORITHM

Given a step size η_t at iteration t , the optimization of DANAM proceeds by iteratively executing the following three steps in a single loop until a suitable convergence criterion is met.

Step 1. Estimate the gradient of the average task loss on a mini-batch b of size m . The gradient is computed as:

$$g_t = \frac{1}{m} \sum_{i \in b} \nabla_{\Theta} \mathcal{L}(\mathbf{x}_i, y_i; \Theta_t), \quad (9)$$

where $\mathcal{L}(\mathbf{x}_i, y_i; \Theta_t)$ is the task dependent loss function.

Step 2. Update a gradient descent step:

$$S_t = \Theta_t - \eta_t g_t, \quad (10)$$

where $S_t := [s_{1,t}, \dots, s_{p,t}]$, and $\Theta_t := [\theta_{1,t}, \dots, \theta_{p,t}]$ denotes the parameter set at iteration t , with $\theta_{j,t} \in \mathbb{R}^{k_j}$ being the parameter vector associated with the j -th subnetwork.

Step 3. Update predictor parameter Θ by applying the group-wise proximal operator of the sparsity regularizer:

$$\theta_{j,t+1} = \left(1 - \frac{\eta_t \lambda \sqrt{k_j}}{\|s_{j,t}\|_2} \right)_+ s_{j,t}, \quad (11)$$

where $(x)_+ = \max(0, x)$. This group soft-thresholding operation is applied across all subnetworks to yield the parameter vector Θ_{t+1} . Theoretically, as stated in [13, 20], this efficient forward-backward optimization strategy could find the minimizer of DANAM.

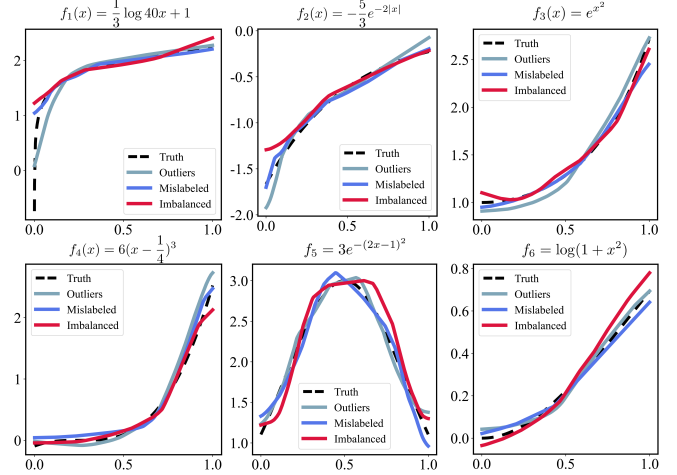


Fig. 2. Curves of the true component functions and their DANAM estimates under three types of data perturbations. The data are generated as $y_i = \text{sign}(\sum_{j=1}^6 f_j^*(\mathbf{x}_i^j))$, where $\mathbf{x}_i \sim \mathcal{U}([0, 1]^p)$. The data are perturbed in three scenarios: 10% outliers; 10% mislabeled samples; and imbalanced class distributions at a 1:10 ratio.

4. EXPERIMENTS

This section validates the effectiveness of DANAM on both simulated and real-world data. All experiments are implemented in Python on a PC with Intel(R) Xeon(R) Platinum 8175M CPU and NVIDIA RTX A6000 GPU.

4.1. Experimental Settings

Baselines. We compare the proposed DANAM with several state-of-the-art models, including MLP, XGBoost, Neural Additive Models (NAM) [1], Neural Basis Models (NBM) [21], and Concept-based Taylor Additive Models (CAT) [8] with second-order concepts. Classification performance in noisy and imbalanced settings is evaluated using Macro-F1 and reported with standard deviations. For ablation analysis, DANAM⁻ implies removing the sparsity-induced penalty. Each experiment is repeated 20 times.

Parameter Selection. For all baselines, the following parameters are selected using the same strategies, i.e., the regularization coefficient $\lambda \in [10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}]$, batch size $b \in [32, 64, 128]$, bandwidth $u \in [0.4, 0.3, 0.2, 0.1]$, learning rates initialized at 10^{-3} with a polynomial decay (power = 0.5), and a maximum iteration count of $T = 10^4$. Other parameters are set according to the corresponding publications. The optimal parameters for each model are selected after a 5-fold cross-validation.

Real Datasets. We conduct experiments on four benchmark classification datasets: Glioma Grading Clinical (Glioma, $n = 839$, $p = 23$), Telco Customer Churn (Telco, $n = 7043$, $p = 21$), Parkinson's Disease (Parkinson, $n =$

Table 1. Experimental results (average Macro-F1 score ($\uparrow$) $\pm$ standard deviation) for classification. Here, r_1 and r_2 denote the rates of noisy labels and category imbalance, respectively. The best and second-best results are highlighted in **bold** and underlined, respectively.

Datasets	r_1	r_2	MLP	XGBoost	NAM	NBM	CAT	DANAM	DANAM ⁻
Glioma	0%	1:1	0.880 ± 0.025	0.875 ± 0.027	0.871 ± 0.022	0.882 ± 0.024	<u>0.885 ± 0.025</u>	0.888 ± 0.013	0.883 ± 0.015
	0%	1:10	0.737 ± 0.101	0.742 ± 0.107	0.739 ± 0.108	0.754 ± 0.103	0.776 ± 0.067	0.806 ± 0.054	0.786 ± 0.062
	10%	1:10	0.722 ± 0.117	0.735 ± 0.119	0.731 ± 0.115	0.743 ± 0.118	0.754 ± 0.106	0.769 ± 0.072	0.759 ± 0.098
Telco	0%	1:1	0.733 ± 0.026	0.748 ± 0.014	0.734 ± 0.023	0.730 ± 0.028	0.757 ± 0.014	<u>0.752 ± 0.010</u>	0.744 ± 0.012
	0%	1:10	0.657 ± 0.089	<u>0.687 ± 0.067</u>	0.654 ± 0.087	0.658 ± 0.074	0.682 ± 0.052	0.701 ± 0.056	0.685 ± 0.062
	10%	1:10	0.621 ± 0.112	<u>0.654 ± 0.095</u>	0.607 ± 0.121	0.629 ± 0.106	0.656 ± 0.093	0.683 ± 0.071	0.671 ± 0.094
Parkinson	0%	1:1	0.829 ± 0.051	0.833 ± 0.045	0.811 ± 0.023	0.844 ± 0.020	<u>0.856 ± 0.015</u>	0.859 ± 0.011	0.852 ± 0.021
	0%	1:10	0.754 ± 0.079	0.749 ± 0.064	0.750 ± 0.077	0.776 ± 0.055	<u>0.788 ± 0.038</u>	0.795 ± 0.032	0.784 ± 0.035
	10%	1:10	0.708 ± 0.108	0.714 ± 0.111	0.711 ± 0.134	0.723 ± 0.090	0.751 ± 0.067	0.763 ± 0.056	0.758 ± 0.064
Ionosphere	0%	1:1	0.948 ± 0.026	0.953 ± 0.021	0.981 ± 0.022	0.985 ± 0.012	0.982 ± 0.018	<u>0.983 ± 0.014</u>	0.979 ± 0.019
	0%	1:10	0.832 ± 0.071	0.847 ± 0.065	0.872 ± 0.034	0.876 ± 0.021	0.881 ± 0.019	0.885 ± 0.017	0.879 ± 0.022
	10%	1:10	0.786 ± 0.092	0.798 ± 0.088	0.816 ± 0.075	0.821 ± 0.073	0.828 ± 0.076	0.839 ± 0.049	<u>0.834 ± 0.068</u>

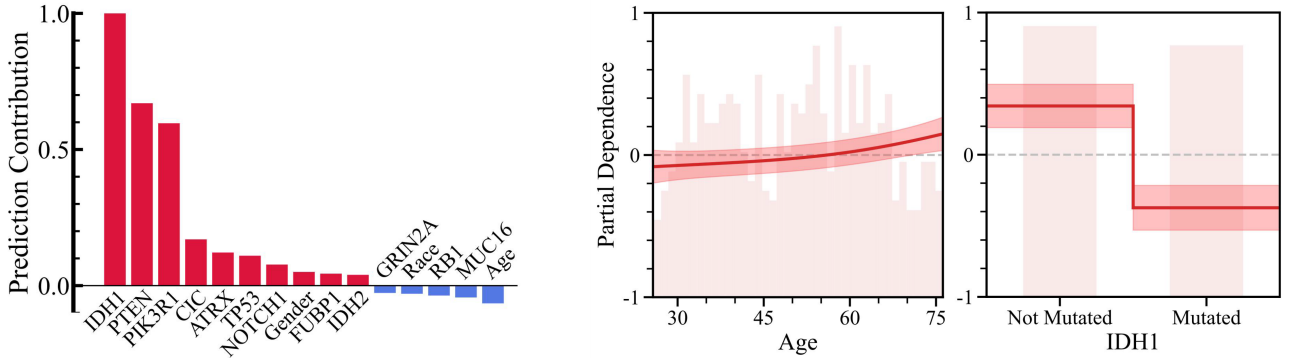


Fig. 3. Experimental results for DANAM. Left panel: Local explanations for the glioma grading task on a sample patient, showing that DANAM selects far fewer features, thereby enhancing interpretability. Right panel: Partial dependence plot for glioma grading with associated epistemic uncertainty ($\pm$ standard deviation).

756, $p = 753$), and Ionosphere ($n = 351$, $p = 34$). Similar to the experimental design in [22], each dataset is randomly split into training, validation, and test sets with an 8:1:1 ratio. Furthermore, a percentage of “ r_1 ” samples from the training set are corrupted by flipping their labels to the next (or first) category for classification. The maximum ratio between the sizes of major and minor classes is denoted as “ r_2 ”.

4.2. Experimental Results

Figure 2 illustrates results on the simulated dataset. The learned subnetworks of DANAM under three different perturbation settings closely approximate the actual component functions, thereby demonstrating both interpretability and precise recovery of individual functions. This robustness is further supported by classification results on real-world datasets in Table 1. Under the challenging conditions with both label noise and severe class imbalance, most baseline models exhibit substantial degradation in Macro-F1, whereas DANAM consistently attains the highest scores across nearly all settings. Although specialized architectures such as CAT occasionally outperform DANAM in clean, balanced cases

(e.g., the Telco dataset with $r_1 = 0\%$, $r_2 = 1:1$) due to their concept-aggregation mechanism, the overall results confirm that DANAM offers superior robustness and accuracy. Furthermore, as shown in Figure 3, DANAM improves model trustworthiness by selecting significantly fewer features for local explanations while also providing a principled quantification of epistemic uncertainty associated with its predictions.

5. CONCLUSION

This paper introduces a novel distribution-aware neural additive model based on maximum likelihood estimation, aiming to achieve robustness and interpretability simultaneously. DANAM attains robustness against complex noise structures while preserving interpretability through component-wise modeling and group sparsity regularization. Extensive experiments demonstrate that DANAM consistently outperforms state-of-the-art baselines. Overall, this work highlights the promise of maximum likelihood estimation for building interpretable neural architectures to handle noisy data in high-stakes domains.

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